

Assignment 04

Individual Assignment

Minimum Expectations:

- You must implement all of the techniques in your .ipynb file that you have learned this semester to demonstrate how much you were able to increase the accuracy and reduce overfitting:
 1. 50 lines of code minimum
 2. Showing the overfitting issue, and at least 2 ways to reduce overfitting based on the topics below.
 3. Showing at least 4 graphs to elaborate on loss, accuracy and overfitting.
- Your code and explanations must include these **topics**:
 1. Data preparation
 2. Train, validation, test set splits
 3. L1, L2, or Dropout Regularization
 4. Overparameterization (too many weights)
 5. Data augmentation
 6. Convnets
 7. Number of epochs required for acceptable metrics

Data

Choose *any* dataset online that you would like to work with. Optionally, you can use one of the datasets below:

- **CIFAR10 DATA:** <https://towardsdatascience.com/getting-started-with-datasets-in-keras-468ee92236ee>
- **Keras Data sets:** https://www.tensorflow.org/api_docs/python/tf/keras/datasets
- **Data Quest:** <https://www.dataquest.io/blog/free-datasets-for-projects/>

Submission Instructions:

- YOU MUST SUBMIT A LINK TO YOUR ONLINE VIDEO (YouTube, Vimeo, etc.) as a COMMENTS SUBMISSION SECTION on your D2L SUBMISSION. Omitting this comment will result in a 0% for your total mark. Video should be between 5-15 minutes. Please remember to make your video unlisted or public, but it cannot be private. Please verify it's accessible using an incognito tab. (5%)
- The **assignment04.ipynb** file that contains all the code. (5%)